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NeuroScan AI : Brain MRI Tumor Segmentatio

1. Project Title

AI-Based Brain Tumor Segmentation System , a two-stage pipeline that detects the presence of a tumor in brain MRI slices and localizes it with pixel-level segmentation, using the BraTS2020 dataset.

2. Team Group Details

Subject: Python for Data Science (PDS) | Activity: Problem-Based Learning (PBL)

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3. Project Description & Objectives

This project uses machine learning and deep learning techniques to analyze brain MRI scans (BraTS2020, FLAIR modality) and detect tumor presence and location. The system operates as a two-stage cascade: a CNN classifier first screens individual 2D slices for tumor presence, and a 2D U-Net then performs pixel-level segmentation only on the slices flagged as containing a tumor — combining outputs into a single visual overlay showing tumor location across the full scan.

Objectives:

- Build a two-stage pipeline that takes brain MRI (FLAIR) slices as input and identifies tumor presence and location.
- Classify individual 2D MRI slices as Tumor / No Tumor using a Convolutional Neural Network (CNN).
- Segment tumor regions at the pixel level using a 2D U-Net architecture with Dice loss, applied only to slices flagged by the classifier (cascade design).
- Preprocess and analyze volumetric NIfTI image data using Python libraries including NumPy, PyTorch, MONAI/nibabel, and Matplotlib.
- Evaluate model performance using accuracy, precision, recall, F1-score (classification) and Dice coefficient (segmentation), with patient-level train/validation/test splitting to prevent data leakage.
- Analyze failure patterns (e.g., correlation between tumor size and segmentation accuracy) to understand model limitations.

4. Dataset to be Used

The primary dataset selected for this project is BraTS2020 (Brain Tumor Segmentation Challenge 2020), accessed via Kaggle.

- Source: Kaggle — brats20-dataset-training-validation (originally from the Multimodal Brain Tumor Segmentation Challenge).
- Size: 369 patients (368 used after excluding one incomplete case), each with 4 MRI modalities (T1, T1ce, T2, FLAIR) and an expert-annotated ground-truth segmentation mask.
- Slices: approximately 50,700 usable 2D FLAIR slices extracted across all patients for classification; ~24,000+ tumor-containing slices used for segmentation training.
- Format: NIfTI (.nii) volumetric format, 240×240×155 voxels per patient.
- Split: patient-level 80/10/10 train/validation/test split (fixed random seed) to ensure no data leakage between training and evaluation.

5. Methodology

The following workflow outlines the step-by-step approach followed, from dataset collection through to the final combined prediction output:

